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Master's thesis for Mr. Yucan Luo, matriculation no. 408454

Development of an Enhanced Semantic Segmentation Method for Large-Scale Point Cloud Processing on Resource-Constrained Devices

Problem:

Point cloud semantic segmentation is a fundamental task for 3D scene understanding. Current state-of-the-art methods achieve high accuracy on benchmark datasets, but face a critical limitation when applied to large-scale scenes.

Scale vs. Resource Conflict: Standard benchmark scenes (e.g., ScanNet with ~10^5 points per scene) can be processed on modern GPUs. However, large-scale scenes present significant challenges:

- Architectural heritage (ArCH): More than 10-100 million points per scene
- Urban mapping (SensatUrban, DALES): hundreds of millions of points
- Outdoor benchmarks (Semantic3D): billions of points total, but cant validation

State-of-the-art methods (both point-based like PTv3 and voxel-based like MinkowskiNet) are optimized for standard-scale scenes. When applied to large-scale scenes, they either run out of memory or require spatial partitioning strategies.

Current Approaches and Limitations:

- Architecture-level efficiency improvements (sparse convolutions, serialized attention) improve memory efficiency but do not fundamentally solve the large-scale processing problem

- Users must implement their own spatial partitioning strategies when processing large scenes

- No systematic study exists on how processing parameters (block size, overlap ratio) affect segmentation performance across different scene types and algorithms

- Parameter selection relies on trial-and-error rather than principled guidelines

Research Opportunity: A systematic, experiment-driven approach is needed to:

1. Establish the relationship between processing parameters, scene characteristics, and segmentation performance through Design of Experiments (DoE)
2. Derive generalizable patterns/formulas that predict optimal parameters for given scenarios
3. Validate the derived models across different datasets and algorithms

Enhancement Strategy Context: Mr. Fan Qinyuan has developed an enhancement strategy for large-scale point cloud semantic segmentation, which includes the following key components:

- **Coordinate scaling and spatial cropping**: Transform point coordinates with a scale factor and crop scenes into fixed-size spatial blocks using a sliding window approach
- **Data augmentation:** Apply geometric transformations including translation jitter, random flipping, scaling, and rotation for training robustness
- **Adaptive down-sampling**: Probabilistic point dropping based on voxel density to balance computational load

The strategy involves multiple configurable parameters:

- `scale`: Coordinate scaling factor (e.g., 50)
- `spatial_shape`: Spatial cropping dimensions (e.g., [512, 512, 512])
- `step`: Cropping step size for sliding window (e.g., 32)
- `jitter_range`: Translation perturbation range for data augmentation (e.g., ±2m)
- `voxel_size`: Voxel down-sampling size range (e.g., 0.01-0.03m)
- `drop_prob`: Instance dropping probability (e.g., 0.1)

Currently, these parameters are manually tuned based on experience. A systematic method for adaptive parameter selection would significantly improve the strategy's applicability across diverse scenes and hardware configurations.

Platform Context: Mr. Fan Qinyuan has also developed the PoLaAll platform, which integrates semantic segmentation for point cloud annotation. A principled parameter selection method enabling large-scale segmentation would significantly expand the platform's applicability.

Two technical questions arise:

1. What is the quantitative relationship between processing parameters (block size, overlap ratio) and enhancement strategy parameters (scale, spatial_shape, step, voxel_size, drop_prob), scene characteristics, and segmentation accuracy?

2. Can generalizable patterns be derived from systematic experiments to guide parameter selection across different scenarios?

Research Objectives

This thesis aims to develop an **adaptive enhancement strategy for large-scale point cloud semantic segmentation** that automatically selects optimal processing parameters based on scene characteristics and hardware constraints.

To achieve this goal, the thesis will:

- Conduct systematic Design of Experiments (DoE) to analyze the relationship between processing parameters, scene characteristics, and segmentation performance across multiple SOTA architectures.

- Derive generalizable patterns or predictive models from experimental data that form the basis of the adaptive strategy.

- Validate the adaptive strategy on diverse benchmark datasets from standard scale to ultra-large scale, demonstrating practical applicability.

Task

Objective 1: Enhancement Method Development

Phase 1: Literature Review and Baseline Analysis

- Review existing large-scale point cloud segmentation methods and their resource requirements

- Analyze point-based SOTA methods (e.g. Point Transformer, PTv3)

- Analyze voxel-based SOTA methods (e.g. MinkowskiNet, SPVCNN)

- Identify computational bottlenecks and memory consumption patterns

Phase 2: Design of Experiments (DoE)

- Define experimental factors based on the enhancement strategy:
  - Scaling and cropping parameters:
    - `scale`: Coordinate scaling factor (range: 20-100)
    - `spatial_shape`: Spatial cropping dimensions (e.g., [256,256,256] to [1024,1024,1024])
    - `step`: Sliding window step size controlling block overlap (range: 16-128)
  - Data augmentation parameters:
    - `jitter_range`: Translation perturbation range (range: 0-5m)
  - Down-sampling parameters:
    - `voxel_size_range`: Voxel down-sampling size range (range: 0.005-0.05m)
    - `drop_prob`: Instance dropping probability (range: 0-0.3)
  - Processing modes:
    - Spatial alignment: with alignment (overlapping regions merged consistently) vs. without alignment (independent block processing)
  - Scene characteristics: point density (points/m³), scene extent, geometric complexity
  - Algorithm type: point-based (PTv3, Point Transformer) vs voxel-based (MinkowskiNet, Cylinder3D)
  - Hardware constraint: available VRAM (8GB, 12GB, 24GB)

- Define response variables:
  - Segmentation quality: mIoU, AP, etc.
  - Resource consumption: peak memory, processing time
- DoE design:
  - Full factorial or fractional factorial design for parameter space exploration
  - Multiple datasets to capture scene variability
  - Multiple runs for statistical significance
- Execute experiments systematically and collect data

Phase 3: Pattern Extraction and Model Derivation

- Analyze experimental data:
  - Correlation analysis between factors and responses
  - Identify significant factors and interactions
  - Visualize parameter-performance relationships
- Derive predictive models/patterns:
  - Regression models: mIoU = f (scale, spatial_shape, step, voxel_size, drop_prob, point_density, algorithm_type, ...)
  - Memory model: M = g (block_size, voxel_size_range, point_count, algorithm_type, ...)
  - Decision rules or lookup tables for practical use
- Model requirements:
  - Interpretable: can explain why certain parameters work better
  - Generalizable: applicable beyond training scenarios
  - Practical: easy to apply given new scene characteristics

Phase 4: Implementation

- Implement adaptive parameter selection module based on derived models:
  - Automatic estimation of optimal scale, spatial_shape, step based on scene characteristics
  - Automatic selection of voxel_size_range and drop_prob based on hardware constraints
- Extend the existing enhancement strategy with adaptive parameter configuration
- Implement scene partitioning pipeline with configurable block size and overlap
- Integrate with point-based SOTA
- Integrate with voxel-based SOTA
- Implement boundary handling and result merging for partitioned processing

Objective 2: Validation and Evaluation

Phase 5: Experimental Setup

- Prepare evaluation datasets across different scales:
  - Large-scale indoor: S3DIS (~10^6 points per area), ArCH (5-15 million points per scene)
  - Large-scale outdoor: SensatUrban (3 billion points), DALES (500 million points, aerial LiDAR)

- Large-scale driving: KITTI-360 (73.7km coverage)

- Define resource-constrained test environments (e.g., 8GB, 12GB VRAM limits)
- Establish baseline performance of SOTA methods under resource constraints

Phase 6: Model Validation

- Validate derived models on held-out scenarios:
  - Datasets not used in DoE
  - Different hardware configurations
- Compare model-predicted parameters vs. empirically optimal parameters
- Measure prediction accuracy

Phase 7: Performance Evaluation

- Compare model-guided parameter selection vs. arbitrary fixed parameters:
  - Show that model-guided selection achieves better or comparable accuracy
  - Show that model-guided selection avoids OOM errors
- Evaluate across scene scales:
  - Large indoor (S3DIS)
  - Large outdoor (ArCH, DALES)
  - Large driving (KITTI-360)
- Metrics:
  - Segmentation quality: mIoU, per-class IoU, AP
  - Resource consumption: peak memory, total processing time

Phase 8: Analysis

- Analyze derived patterns:
  - Which factors most strongly affect performance?
  - Are there interaction effects between parameters?
  - How do patterns differ between point-based and voxel-based methods?
- Spatial alignment analysis:
  - Quantify accuracy improvement from spatial alignment vs. non-alignment
  - Analyze alignment overhead (time, memory) vs. accuracy gain
  - Identify scenarios where alignment is critical vs. optional
- Generalization analysis:
  - Do patterns transfer across scene types (indoor vs. outdoor)?
  - Do patterns transfer across algorithms?
- Document limitations and applicable scope

Worksteps

1. Literature Review

   - Survey large-scale point cloud segmentation methods
   - Study DoE methodology and experimental design principles

2. Incremental Development

- Phase 1: Literature review and baseline analysis
- Phase 2-3: DoE execution and pattern derivation
- Phase 4: Implementation
- Phase 5-8: Validation and analysis

3. Experimentation

- Systematic DoE experiments across datasets and algorithms
- Model validation on held-out scenarios
- Performance comparison studies

4. Documentation

- Technical documentation of derived models/patterns
- Experiment results and statistical analysis
- Record demonstration videos

Expected Outcomes

1. Adaptive Enhancement Strategy: A principled method for automatic parameter selection in large-scale point cloud segmentation, based on scene characteristics and hardware constraints

2. DoE-Derived Patterns: Experimentally validated formulas or rules describing parameter-performance relationships

3. SOTA Integration: Implementation compatible with both point-based and voxel-based architectures

4. Benchmark Results: Evaluation across large-scale datasets (S3DIS, ArCH, DALES, KITTI-360), demonstrating strategy effectiveness and generalizability

Organization

This research develops resource-efficient semantic segmentation methods for large-scale point clouds, advancing the practical deployment of perception algorithms on constrained hardware. The results of the literature review, method design, and experimental investigations are presented in clear form (tables, graphs) and discussed from a scientific perspective. The results are presented as part of the final measurement engineering seminar, publicly at university.

The contact person for content-related advice at the department MDT is Mr. Qinyuan Fan.

With kind regards

Prof. Dr.-Ing. Clemens Gühmann